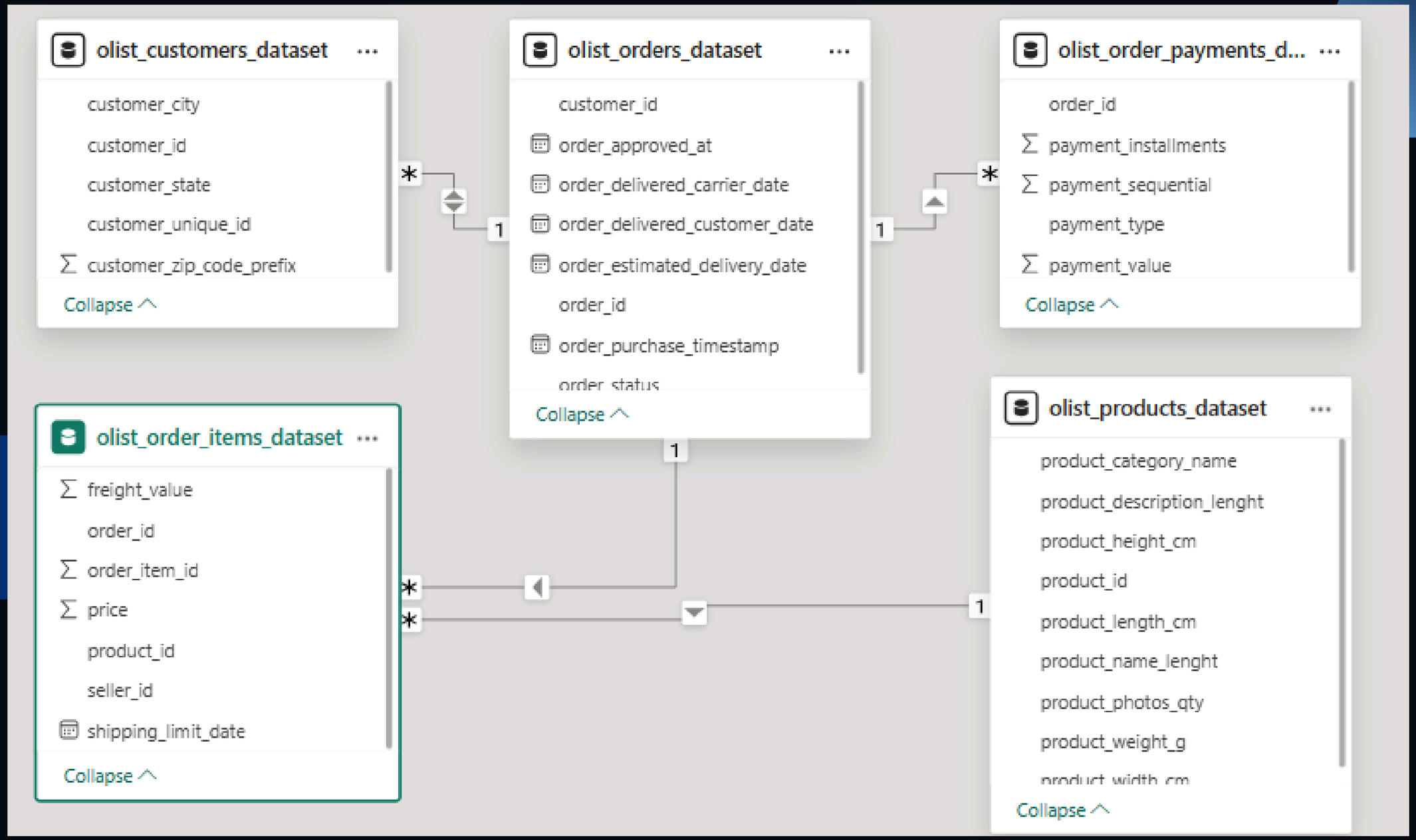
OLIST E-COMMERCE PRESENTATION

ABOUT

This project analyzes the Olist e-commerce dataset using SQL to derive meaningful business insights. It focuses on key metrics such as total revenue, customer behavior, sales trends, and product performance. The analysis helps identify patterns in customer retention, payment preferences, and category-wise sales to support data-driven decision-making.

MODEL


olist_customers_dataset
customer_city
customer_id
customer_state
customer_unique_id
customer_zip_code_prefix
Collapse
olist_orders_dataset
customer_id
order_approved_at
order_delivered_carrier_date
order_delivered_customer_date
order_estimated_delivery_date
order_id
order_purchase_timestamp
order_status
Collapse
olist_order_payments_d...
order_id
payment_installments
payment_sequential
payment_type
payment_value
Collapse
olist_order_items_dataset
freight_value
order_id
order_item_id
price
product_id
seller_id
shipping_limit_date
Collapse
olist_products_dataset
product_category_name
product_description_lenght
product_height_cm
product_id
product_length_cm
product_name_lenght
product_photos_qty
product_weight_g
product_width_cm
Collapse
*
1
*
1
1
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*
1

REFERENCES

Below mentioned are the tables along with its description that contributed to fetch the records

Olist_customers_dataset - This table contains customer_id, state and city from which a customer belongs along with its zip code For every new transaction, it also includes generated new customer_unique_id.

Olist_orders_dataset - It highlights details related to orders like customer_id who purshased an order, order_id generated at every new order, order_status and order approved, carrier and delivery date.

Olist_order_items_dataset - It includes order_id, order_item_id, prodcut_id, seller_id, shopping_limit_date, price and freight_value.

Olist_products_dataset - It decribes product dimensions like product's name length, description length, photos qty, weight (in gms), length (in cms), height (in cms), category name and product id.

Olist_order_payments_dataset - It includes rows like order_id, payment_sequential, payment_type, payment_installments and payment_value.

TOTAL NUMBER OF ORDERS

```sql
SELECT
    COUNT(order_id) AS Total_number_of_orders
FROM
    olist_orders_dataset;
```

Result Grid | Filter Rows:

	Total_number_of_orders
▶	99441

TOTAL NUMBER OF CUSTOMERS

```sql
SELECT
    COUNT(DISTINCT customer_unique_id) AS Total_number_of_customers
FROM
    olist_customers_dataset;
```

Result Grid | Filter Rows:

	Total_number_of_customers
▶	96096

TOTAL REVENUE GENERATED

```sql
SELECT
    ROUND(SUM(price + freight_value), 2) AS Total_revenue_generated
FROM
    olist_order_items_dataset;
```

Result Grid | Filter Rows:

	Total_revenue_generated
▶	15843553.24

AVERAGE ORDER VALUE

```sql
SELECT
    ROUND(SUM(price + freight_value) / COUNT(DISTINCT order_id),
            2) AS Average_order_value
FROM
    olist_order_items_dataset;
```

Result Grid | Filter Rows:

	Average_order_value
▶	160.58

MONTHLY SALES TREND

```sql
SELECT
    DATE_FORMAT(order_purchase_timestamp, '%Y-%M') AS Month,
    ROUND(SUM(price + freight_value), 2) AS Total_sales
FROM
    olist_order_items_dataset otd
        JOIN
    olist_orders_dataset od ON otd.order_id = od.order_id
GROUP BY DATE_FORMAT(order_purchase_timestamp, '%Y-%M')
ORDER BY month;
```

Result Grid | Filter Rows:

Month	Total_sales
2016-December	19.62
2016-October	56808.84
2016-September	354.75
2017-April	412422.24
2017-August	668204.6
2017-December	863547.23
2017-February	286280.62
2017-January	137188.49
2017-July	584971.62
2017-June	502963.04
2017-March	432048.59
2017-May	586190.95
2017-November	1179143.77
2017-October	769312.37
2017-September	720398.91
2018-April	1159698.04
2018-August	1003308.47
2018-February	986908.96
2018-January	1107301.89

TOP 10 PRODUCTS BY REVENUE

```sql
SELECT
    pd.product_id,
    ROUND(SUM(otd.price + otd.freight_value), 2) AS Total_sales
FROM
    olist_order_items_dataset otd
        JOIN
    olist_products_dataset pd ON otd.product_id = pd.product_id
GROUP BY pd.product_id
ORDER BY Total_sales DESC
LIMIT 10;
```

Result Grid | Filter Rows:

product_id	Total_sales
bb50f2e236e5eea0100680137654686c	67606.1
d1c427060a0f73f6b889a5c7c61f2ac4	60976.03
6cdd53843498f92890544667809f1595	59093.99
99a4788cb24856965c36a24e339b6058	51071.6
d6160fb7873f184099d9bc95e30376af	50326.18
3dd2a17168ec895c781a9191c1e95ad7	48212.22
aca2eb7d00ea1a7b8ebd4e68314663af	44820.76
5f504b3a1c75b73d6151be81eb05bdc9	41725.81
25c38557cf793876c5abdd5931f922db	40311.95
53b36df67ebb7c41585e8d54d6772e08	39957.93

TOP 5 CUSTOMERS BY SPENDING

```sql
SELECT
    cd.customer_unique_id,
    ROUND(SUM(otd.price + otd.freight_value), 2) AS Total_sales
FROM
    olist_customers_dataset cd
        JOIN
    olist_orders_dataset od ON cd.customer_id = od.customer_id
        JOIN
    olist_order_items_dataset otd ON od.order_id = otd.order_id
GROUP BY cd.customer_unique_id
ORDER BY Total_sales DESC
LIMIT 5;
```

Result Grid | Filter Rows:

customer_unique_id	Total_sales
0a0a92112bd4c708ca5fde585afaa872	13664.08
da122df9eeddfedc1dc1f5349a1a690c	7571.63
763c8b1c9c68a0229c42c9fc6f662b93	7274.88
dc4802a71eae9be1dd28f5d788ceb526	6929.31
459bef486812aa25204be022145caa62	6922.21

ORDERS BY PAYMENT METHOD

```sql
SELECT
    opm.payment_type,
    COUNT(DISTINCT opm.order_id) AS No_of_orders
FROM
    olist_order_payments_dataset opm
GROUP BY opm.payment_type;
```

Result Grid | Filter Rows:

payment_type	No_of_orders
boleto	19784
credit_card	76505
debit_card	1528
not_defined	3
voucher	3866

CATEGORY-WISE SALES PERFORMANCE

```sql
SELECT
    opd.product_category_name,
    ROUND(SUM(otd.price + otd.freight_value), 2) AS Total_sales
FROM
    olist_products_dataset opd
        JOIN
    olist_order_items_dataset otd ON opd.product_id = otd.product_id
GROUP BY opd.product_category_name
ORDER BY Total_sales DESC;
```

Result Grid | Filter Rows:

product_category_name	Total_sales
beleza_saude	1441248.07
relogios_presentes	1305541.61
cama_mesa_banho	1241681.72
esporte_lazer	1156656.48
informatica_acessorios	1059272.4
moveis_decoracao	902511.79
utilidades_domesticas	778397.77
cool_stuff	719329.95
automotivo	685384.32
ferramentas_jardim	584219.21
brinquedos	561372.55
bebes	478157
perfumaria	453338.71
telefonia	394883.32
moveis_escritorio	342532.65
papelaria	277741.71
pet_shop	253876.65
pcs	232799.43
instrumentos_musicais	210137.37

AVERAGE ITEMS PER ORDER

```sql
SELECT
    ROUND(AVG(items_per_order), 2) AS Avg_items_per_order
FROM
    (SELECT
        otd.order_id, COUNT(*) AS items_per_order
    FROM
        olist_order_items_dataset otd
    GROUP BY otd.order_id) t;
```

Result Grid | Filter Ro

	Avg_items_per_order
▶	1.14

REPEAT VS NEW CUSTOMERS

```sql
SELECT
    CASE
        WHEN order_count = 1 THEN 'One_time'
        ELSE 'Repeat'
    END AS customer_type,
    COUNT(customer_unique_id) AS Total_customers
FROM
    (SELECT
        c.Customer_unique_id, COUNT(o.order_id) AS order_count
    FROM
        olist_orders_dataset o
    JOIN olist_customers_dataset c ON o.customer_id = c.customer_id
    GROUP BY c.customer_unique_id) t
GROUP BY customer_type;
```

customer_type	Total_customers
One_time	93099
Repeat	2997

CUSTOMER LIFETIME VALUE (CLV – BASIC)

```sql
SELECT
    cd.customer_unique_id,
    COUNT(DISTINCT od.order_id) AS Total_orders,
    ROUND(SUM(price + freight_value), 2) AS Total_sales
FROM
    olist_orders_dataset od
        JOIN
    olist_customers_dataset cd ON cd.customer_id = od.customer_id
        JOIN
    olist_order_items_dataset otd ON od.order_id = otd.order_id
GROUP BY cd.customer_unique_id;
```

customer_unique_id	Total_orders	Total_sales
0000366f3b9a7992bf8c76cfdf3221e2	1	141.9
0000b849f77a49e4a4ce2b2a4ca5be3f	1	27.19
0000f46a3911fa3c0805444483337064	1	86.22
0000f6ccb0745a6a4b88665a16c9f078	1	43.62
0004aac84e0df4da2b147fca70cf8255	1	196.89
0004bd2a26a76fe21f786e4fbd80607f	1	166.98
00050ab1314c0e55a6ca13cf7181fecf	1	35.38
00053a61a98854899e70ed204dd4bafe	1	419.18
0005e1862207bf6ccc02e4228effd9a0	1	150.12
0005ef4cd20d2893f0d9fbd94d3c0d97	1	129.76
0006fdc98a402fceb4eb0ee528f6a8d4	1	29
00082cbe03e478190aadbea78542e933	1	126.26
00090324bbad0e9342388303bb71ba0a	1	63.66
000949456b182f53c18b68d6babc79c1	1	82.05
000a5ad9c4601d2bbdd9ed765d5213b3	1	91.28
000bfa1d2f1a41876493be685390d6d3	1	46.85
000c8bdb58a29e7115cfc257230fb21b	1	29
000d460961d6dbfa3ec6c9f5805769e1	1	36.68
000de6019bb59f34c099a907c151d855	1	257.44

TOP-SELLING PRODUCT IN EACH CATEGORY

```
Select * from
(select otd.product_id, pd.product_category_name, count(*) as Total_sales,
Row_number() Over (partition by pd.product_category_name
order by count(*) desc) as rn
from olist_order_items_dataset otd
join olist_products_dataset pd on otd.product_id=pd.product_id
group by pd.product_category_name, otd.product_id) t
where rn=1;
```

product_id	product_category_name	Total_sales	rn
11250b0d4b709fee92441c5f34122aed	agro_industria_e_comercio	22	1
89321f94e35fc6d7903d36f74e351d40	alimentos	119	1
84f5c4f480ad6c9998d6a6860f1a2e41	alimentos_bebidas	24	1
4fe644d766c7566dbc46fb851363cb3b	artes	107	1
b9976e9c22fb1540bd71d1bcd2989475	artes_e_artesanato	5	1
4a7056788ce9287c76ade358f2e2572a	artigos_de_festas	8	1
40e8b425d1a26e2d9cb77363523e05ce	artigos_de_natal	22	1
db5efde3ad0cc579b130d71c4b2db522	audio	48	1
4fcb3d9a5f4871e8362dfedbdb02b064	automotivo	89	1
cac9e5692471a0700418aa3400b9b2b1	bebes	89	1
cd48f265a63e13b762601f5f794c5fca	bebidas	93	1
154e7e31ebfa092203795c972e5804a6	beleza_saude	281	1
880be32f4db1d9f6e2bec38fb6ac23ab	brinquedos	99	1
99a4788cb24856965c36a24e339b6058	cama_mesa_banho	488	1
35afc973633aaeb6b877ff57b2793310	casa_conforto	165	1
ec5b3c8bb77ad22278f6e9d719a83de3	casa_conforto_2	26	1
a39cc58c1b5926b6f9f378daa89f1315	casa_construcao	35	1
1dceebcc5f23c02ea23e16d5bedca000	cds_dvds_musicais	14	1
e988092a8afe5d9393a5c3a7ca17691c	cine_foto	12	1

RANK CUSTOMERS BASED ON SPENDING (RANK())

```
Select cd.customer_unique_id, Round(sum(Price+Freight_value),2) As Total_spending,
Rank() over(order by sum(Price+Freight_value) desc) as customer_rank
from olist_customers_dataset cd
join olist_orders_dataset od on od.customer_id=cd.customer_id
join olist_order_items_dataset otd on otd.order_id=od.order_id
group by cd.customer_unique_id;
```

customer_unique_id	Total_spending	customer_rank
48e1ac109decbb87765a3eade6854098	4681.78	10
c8460e4251689ba205045f3ea17884a1	4655.88	11
edde2314c6c30e864a128ac95d6b2112	4513.32	12
a229eba70ec1c2abef51f04987deb7a5	4445.5	13
edf81e1f3070b9dac83ec83dacdbb9bc	4194.76	14
fa562ef24d41361e476e748681810e1e	4175.26	15
ca27f3dac28fb1063faddd424c9d95fa	4163.51	16
055ec572ac7f3c7bdd04a183830ebe59	4053.08	17
5e713be0853d8986528d7869a0811d2b	4042.74	18
58483a1c055dfb600f57c5b867174542	4034.44	19
011875f0176909c5cf0b14a9138bb691	4016.91	20
f0767ae738c3d90e7b737d7b8b8bb4d1	3979.55	21
bc5e25094a7d51b6aee35236572e64f4	3826.8	22
3fec1a05177762f279472253113d568a	3792.59	23
5d09b0d82126457e2a8ebfb9c9a1ffc4	3736.22	24
931eabdf0636b8fd60369a8d759917d6	3666.42	25
03796b63235e0e0a299084988c662c7e	3602.47	26
59d66d72939bc9497e19d89c61a96d5f	3559.99	27
2c3b08cf3584d8c0a81d9618a73aa219	3526.46	28

RUNNING TOTAL OF REVENUE (WINDOW FUNCTION)

```
Select otd.order_id, od.order_purchase_timestamp,
sum(Price+Freight_value) As Total_revenue,
sum(sum(Price+Freight_value)) over (order by od.order_purchase_timestamp) as running_total
from olist_order_items_dataset otd
join olist_orders_dataset od on od.order_id=otd.order_id
group by od.order_purchase_timestamp, otd.order_id
order by od.order_purchase_timestamp;
```

order_id	order_purchase_timestamp	Total_revenue	running_total
2e7a8482f6fb09756ca50c10d7bfc047	2016-09-04 21:15:19	136.23	136.23
e5fa5a7210941f7d56d0208e4e071d35	2016-09-05 00:15:34	75.06	211.29
bfbd0f9bdef84302105ad712db648a6c	2016-09-15 12:16:38	143.46	354.75
71303d7e93b399f5bcd537d124c0bcfa	2016-10-02 22:07:52	109.34	464.09000000000003
3b697a20d9e427646d92567910af6d57	2016-10-03 09:44:50	45.46	509.55
be5bc2f0da14d8071e2d45451ad119d9	2016-10-03 16:56:50	39.09	548.64
65d1e226dfaeb8cdc42f665422522d14	2016-10-03 21:01:41	35.61	584.25
a41c8759fbe7aab36ea07e038b2d4465	2016-10-03 21:13:36	53.730000000000004	637.98
d207cc272675637bfed0062edffd0818	2016-10-03 22:06:03	133.46	771.44
cd3b8574c82b42fc8129f6d502690c3e	2016-10-03 22:31:31	40.95	812.3900000000001
ae8a60e4b03c5a4ba9ca0672c164b181	2016-10-03 22:44:10	154.57	966.96
ef1b29b591d31d57c0d7337460dd83c9	2016-10-03 22:51:30	92.27000000000001	1059.23
0a0837a5eee9e7a9ce2b1fa831944d27	2016-10-04 09:06:10	65.5	1124.73
1ff217aa612f6cd7c4255c9bfe931c8b	2016-10-04 09:16:33	44.230000000000004	1168.96
ed8c7b1b3eb256c70ce0c74231e1da88	2016-10-04 09:59:03	114.77000000000001	1283.73
c3d9e402b6a0fbe2a5f7fc5b41117c38	2016-10-04 10:16:04	237.45	1521.18
46046adea0e222a29259bad3d922fee8	2016-10-04 10:41:17	214.98	1736.16
80606b26965c5ed21e85a085e0667b63	2016-10-04 11:03:14	178.11	1914.27
79ffdd52a918bbe867895a4b183d6457	2016-10-04 12:06:11	319	2233.27

PEAK ORDER MONTH/DAY

```
Select * from
(Select date(order_purchase_timestamp) as Order_date,
count(od.order_id) as Total_orders,
Rank () over (order by count(od.order_id) desc) as rnk
from olist_orders_dataset od
group by Order_date) t
where rnk=1;
```

	Order_date	Total_orders	rnk
▶	2017-11-24	1176	1

TOP REVENUE-GENERATING CATEGORY

```
Select opd.product_category_name, Round(sum(price+freight_value),2) as Total_sales,
Rank () over (order by Round(sum(price+freight_value),2) desc) as rnk
from olist_products_dataset opd
join olist_order_items_dataset otd on otd.product_id=opd.product_id
group by opd.product_category_name
order by Total_sales desc;
```

product_category_name	Total_sales	rnk
beleza_saude	1441248.07	1
relogios_presentes	1305541.61	2
cama_mesa_banho	1241681.72	3
esporte_lazer	1156656.48	4
informatica_acessorios	1059272.4	5
moveis_decoracao	902511.79	6
utilidades_domesticas	778397.77	7
cool_stuff	719329.95	8
automotivo	685384.32	9
ferramentas_jardim	584219.21	10
brinquedos	561372.55	11
bebes	478157	12
perfumaria	453338.71	13
telefonia	394883.32	14
moveis_escritorio	342532.65	15
papelaria	277741.71	16
pet_shop	253876.65	17
pcs	232799.43	18
instrumentos_musicais	210137.37	19

CUSTOMER RETENTION RATE

```sql
SELECT
    ROUND(COUNT(DISTINCT CASE
                    WHEN order_count > 1 THEN customer_unique_id
                END) * 100.0 / COUNT(DISTINCT customer_unique_id),
            2) AS Retention_rate
FROM
    (SELECT
        cd.customer_unique_id, COUNT(od.order_id) AS order_count
    FROM
        olist_orders_dataset od
    JOIN olist_customers_dataset cd ON cd.customer_id = od.customer_id
    GROUP BY cd.customer_unique_id) t;
```

	Retention_rate
▶	3.12

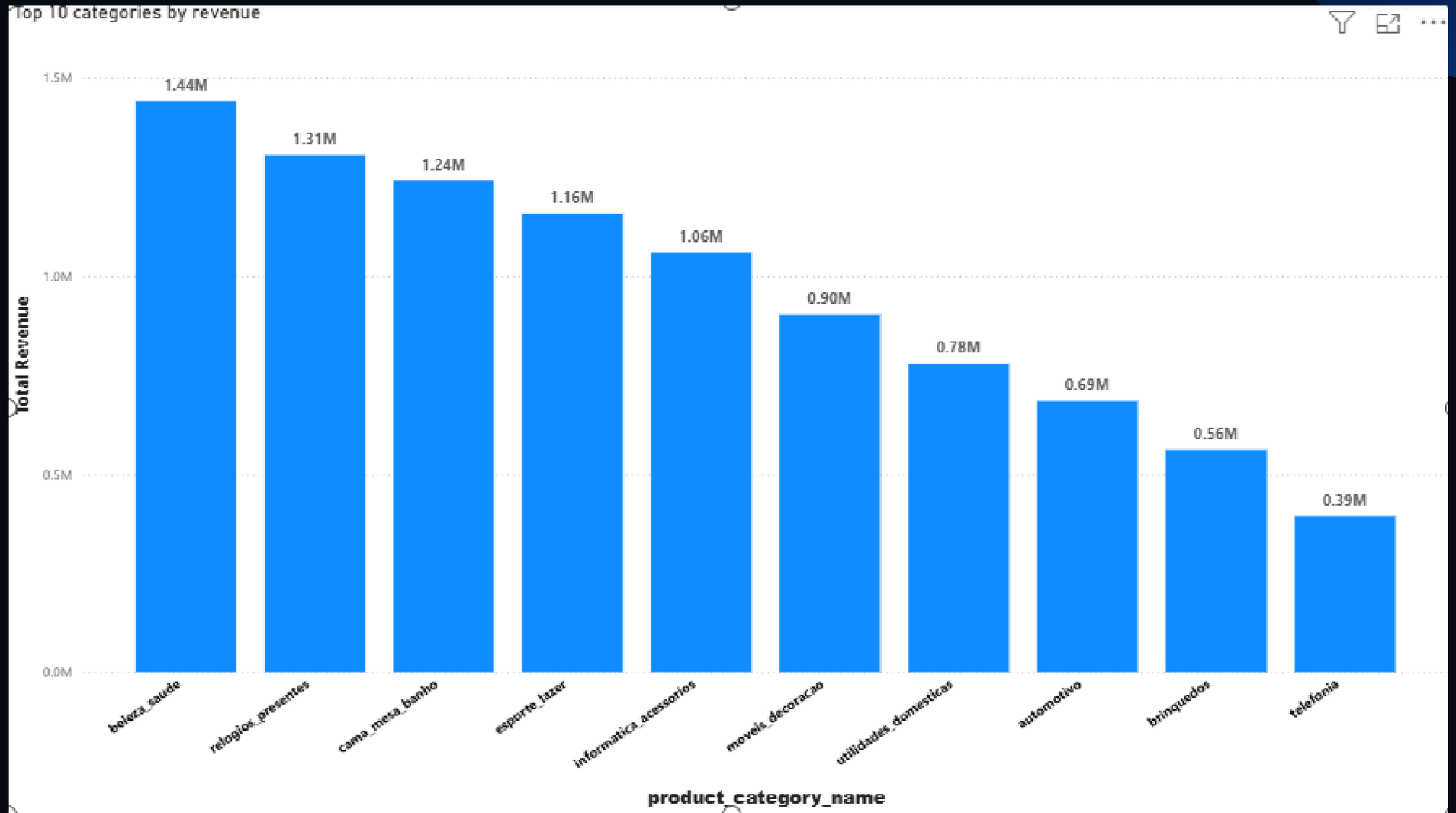
REVENUE BY PAYMENT TYPE VS ORDER COUNT

```sql
SELECT
    opd.payment_type,
    ROUND(SUM(oi.order_revenue), 2) AS total_revenue,
    COUNT(DISTINCT opd.order_id) AS order_count
FROM
    olist_order_payments_dataset opd
        JOIN
    (SELECT
        order_id, SUM(price + freight_value) AS order_revenue
    FROM
        olist_order_items_dataset
    GROUP BY order_id) oi ON oi.order_id = opd.order_id
GROUP BY opd.payment_type
ORDER BY total_revenue DESC;
```

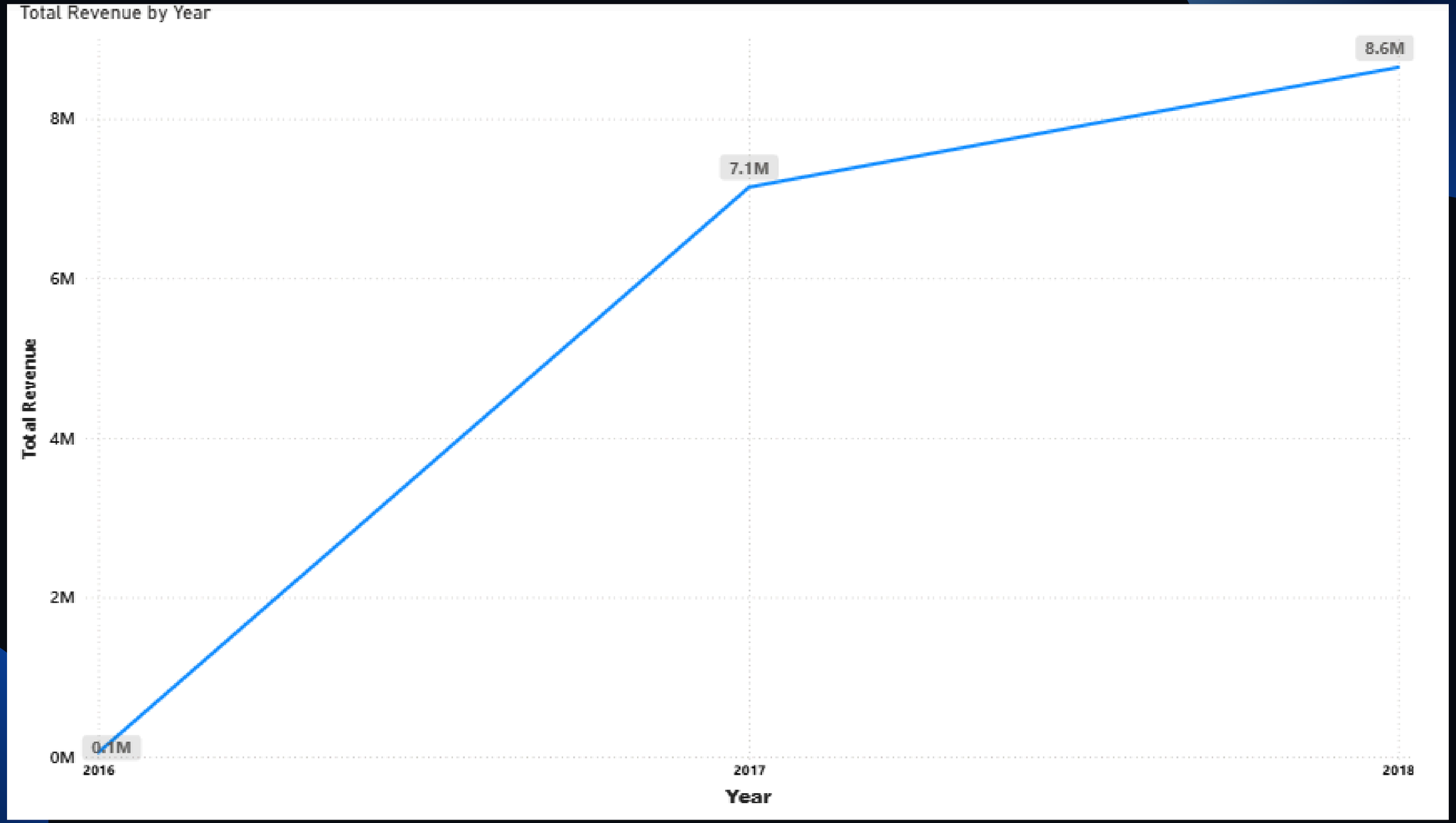
payment_type	total_revenue	order_count
credit_card	12723914.88	75991
boleto	2842240.29	19614
voucher	785081.94	3766
debit_card	215306.74	1521

BI Dashboards

TOP 10 CATEGORIES BY REVENUE



TOTAL REVENUE BY YEAR



KEY INSIGHTS

- ~99K orders generated ~15.8M in total revenue across ~96K customers
- November 2017 recorded peak sales, likely due to seasonal demand (e.g., sale events)
- “Beleza_saude” is the top revenue-generating category
- Boleto is the most used payment method, indicating preference for offline/alternative payments
- Customer retention is very low (~3%), highlighting a strong dependency on new customers

THANK YOU